

UNIT 4

→ Classical free electron theory

• Assumptions

1. Metals contain a large no. of free e^-
2. Electrons move randomly like molecules of ideal gas
3. e^- collisions are instantaneous & random
4. Between collision e^- moves freely under the influence of electric field
5. Avg time b/w collisions $\Rightarrow$ relaxation time (τ)

• Results

$$v_d = -\frac{eE\tau}{m} \quad I = neAv_d \quad \sigma = \frac{ne^2\tau}{m} \quad \rho = \frac{1}{\sigma}$$

explains Ohm's law, electric & thermal conductivity

• Failures

It could not explain

1. low specific heat of metals
2. temp dependence of conductivity
3. why only e^- conduct
4. failed at low temp
5. failed to explain quantum effects
6. failed to give correct value of Lorenz number.

→ Quantum Free electron Theory

• Assumptions

1. e^- obeys Pauli exclusion principle
2. e^- obeys fermi-dirac statistics
3. Allowed e^- energies are quantised
4. e^- confined in 3D ∞ potential box
5. e^- motion is wave like nature

• Failures

Could not explain band gap

2. Fails for semiconductors & insulators

Higher fermi level = better

→ Fermi levels

conductivity

It's the highest occupied energy level of e^- in a metal at absolute 0 temp. At $T = 0\text{ K}$ all energy states below fermi level are completely filled and those above it are empty.

* e^- close to E_F participate in conduction

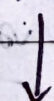
→ Fermi energy (E_F)

$$E_F = \frac{h^2}{8m} \left(\frac{3n}{\pi} \right)^{2/3}$$

$m \rightarrow$ mass of e^-

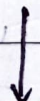
$n \rightarrow$ no of free e^- per unit volume

$$T_F = \frac{E_F}{K_B}$$



Fermi Temp.

$$V_F = \sqrt{\frac{2E_F}{m}}$$



Fermi Velocity

→ Fermi Distribution Function

It gives the probability that an energy state of energy E is occupied by an e^- at a temp. T

$$f(E) = \frac{1}{e^{(E-E_F)/k_B T} + 1}$$

E = energy level E_F = Fermi level

k_B = Boltzmann constant T = absolute temp.

- $f(E)$ ranges from 0 → 1
- 1. $f(E) = 1$ ⇒ state is fully occupied
- 2. $f(E) = 0$ ⇒ state is empty
- 3. $f(E) = 0.5$ at $E = E_F$

→ It explains

- 1. electrical conduction
- 2. why only e^- near E_F contribute
- 3. foundation of band theory

→ Density of states $g(E)$

It's defined as no. of available electron states per unit energy interval per unit volume at energy E

It explains how many quantum states exist at a particular energy level that electron can occupy.

consider electrons in a cube of side L
allowed wave vectors form a lattice
in k -space

→ volume of sphere in radius k of k space
is $\frac{4\pi k^3}{3}$

each state occupies volume in k space
equal to $\frac{(2\pi)^3}{V}$

with 2 spin states

$$N(k) = 2 \cdot \frac{4\pi k^3}{3} \cdot \frac{V}{(2\pi)^3}$$

$$\Rightarrow N(k) = \frac{V k^3}{3\pi^2}$$

also,

$$E = \frac{\hbar^2 k^2}{2m} \Rightarrow k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\Rightarrow dk = \frac{m}{\hbar^2 k} dE$$

$$g(E) = \frac{dN}{V dE}$$

$$\Rightarrow g(E) = \frac{1}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \sqrt{E}$$

$$g(E) \propto \sqrt{E}$$

- _/_/_
- wave function in periodic potential
In a crystalline solid, atoms are arranged in a periodic lattice therefore potential energy of e^- is also periodic.

$$V(x+a) = V(x) \quad a = \text{lattice constant}$$

- effective mass & velocity of electrons
→ In crystal, electrons move under periodic potential of lattice & not behave like free electrons. Their energy depends on the wave vector

$$E = E(k)$$

- The velocity is given by group velocity

$$v = \frac{1}{\hbar} \frac{dE}{dk}$$

- slope of $E-k$ curve gives e^- velocity

If external force F acts

$$\hbar \frac{dk}{dt} = F$$

$$a = \frac{dv}{dt} = \frac{1}{\hbar} \frac{d}{dk} \left(\frac{dE}{dk} \right) \frac{dk}{dt}$$

$$a = \frac{1}{\hbar^2} \frac{d^2E}{dk^2} F$$

$$F = m^* a$$

$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{d^2E}{dk^2}$$

$$\Rightarrow m^* = \frac{\hbar^2}{d^2E/dk^2}$$